Computational Physics I 3630/5630 Fall 2026

Homework Assignment 4

Due: Monday, September 28

1. Markov-chain Monte Carlo and the Metropolis–Hastings algorithm.

Suppose we want to generate configurations X distributed according to a target probability density $\pi(X)$. In a Markov-chain Monte Carlo (MCMC) method, a candidate configuration Y is first generated from the current configuration X according to an *a priori* or *proposal* probability $A(X \to Y)$. In this sense, $A(X \to Y)$ itself may be viewed as a convenient proposal Markov process. The acceptance step then "regularizes" this proposal process so that the resulting Markov chain has the desired stationary distribution $\pi(X)$. The full transition probability is

$$P(X \to Y) = A(X \to Y)\, P_{\rm acc}(X \to Y) \tag{1}$$

$$= A(X \to Y) \min\left[1, \frac{\pi(Y) A(Y \to X)}{\pi(X) A(X \to Y)}\right]. \tag{2}$$

This is the Metropolis–Hastings formula. For a symmetric proposal, $A(X \to Y) = A(Y \to X)$, the proposal probabilities cancel and only the ratio $\pi(Y)/\pi(X)$ is needed. Notice also that the normalization constant of π cancels, so the target distribution need only be known up to an overall multiplicative constant.

We now apply the above MCMC method to sample a one-dimensional target PDF $\pi(x)$. Use a one-dimensional random-walk proposal $y = x + \ell$, where the step ℓ is drawn from a symmetric distribution $\pi_{\rm step}(\ell) = \pi_{\rm step}(-\ell)$. You may choose any step-size distribution that you know how to sample. If you do not want to spend additional time designing the proposal, simply use the uniform choice $\ell \sim U(-\Delta, \Delta)$. For any such symmetric proposal, $A(x \to y) = A(y \to x)$ and the acceptance probability reduces to $\min[1, \pi(y)/\pi(x)]$. A convenient implementation is

procedure update-walker

input: current position x

1. Draw a step ℓ from your chosen symmetric $\pi_{\rm step}(\ell)$ and set $y = x + \ell$.
2. Evaluate the target PDF at the current and proposed positions.
3. Draw a random number r uniformly from $[0, 1]$.
4. If $r < \min[1, \pi(y)/\pi(x)]$, accept and set $x \leftarrow y$; otherwise keep the old x.

return: updated position x

Apply the algorithm to the double-well probability density

$$\boxed{\pi(x) = C \exp\left[-4(x^2-1)^2\right], \qquad -\infty < x < \infty,} \tag{3}$$

with

$$C = \left[\int_{-\infty}^{\infty} e^{-4(x^2-1)^2}\, {\rm d}x\right]^{-1} \simeq 1.05503. \tag{4}$$

Thus the distribution has two peaks near $x = \pm 1$ and a low-probability barrier region near $x = 0$.

Run a sufficiently long Markov chain, discard an initial equilibration portion, and make a normalized histogram of the sampled positions. For plotting, use a finite range such as $-2 \le x \le 2$; do *not* impose this range as an artificial boundary on the random walk. Overlay the analytical curve in Eq. (3). Choose a reasonable width or characteristic step size for your $\pi_{\rm step}(\ell)$ and briefly discuss how the behavior changes if typical steps are made much smaller or much larger (acceptance rate, motion of the walker, and crossing between the two wells).

In-class checkpoint. Before the end of today's coding session, submit a preliminary normalized histogram for Eq. (3), together with the analytical curve, showing that your Metropolis code is working.

2. Metropolis sampling: two interacting particles in a box.

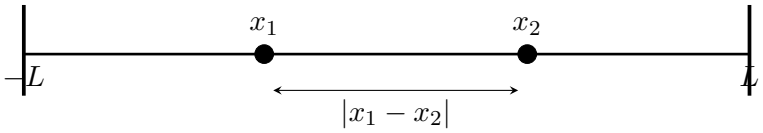
Consider two identical charged particles constrained to move in a one-dimensional box. Their positions are x_1 and x_2, and the total potential energy is

$$E(x_1, x_2) = V(x_1) + V(x_2) + U(x_1 - x_2), \tag{5}$$

where

$$V(x) = \begin{cases} 0, & -L < x < L, \\ \infty, & \text{otherwise}, \end{cases} \qquad U(y) = \frac{e^2}{|y|}. \tag{6}$$

Here the Coulomb prefactor is absorbed into e^2.



At temperature T, the equilibrium configurational probability density is

$$\pi(x_1, x_2) = \frac{1}{Z} e^{-\beta E(x_1, x_2)}, \qquad \beta = \frac{1}{k_B T}. \tag{7}$$

Inside the box,

$$\pi(x_1, x_2) \propto \exp\left[-\frac{\beta e^2}{|x_1 - x_2|} \right]. \tag{8}$$

Introduce the dimensionless interaction strength

$$\lambda = \frac{\beta e^2}{L}, \tag{9}$$

and take $L = 1$ throughout this problem.

You sampled this same distribution previously using rejection sampling. Now sample it using the Metropolis algorithm. For example, from the current configuration $X = (x_1, x_2)$ you may propose

$$Y = (x_1 + \delta_1,\ x_2 + \delta_2), \qquad \delta_1, \delta_2 \sim U(-\Delta, \Delta), \tag{10}$$

which is a symmetric proposal. If either proposed coordinate lies outside the box, the target probability is zero and the move is rejected. You may instead update one particle at a time, provided you clearly state your update rule.

Complete the following tasks:

(i) For $\lambda = 1$, generate equilibrium samples and make a two-dimensional normalized histogram (probability-density heat map) in the (x_1, x_2) plane. **Benchmark your heat map against the result you previously obtained using rejection sampling.** The two methods should produce the same equilibrium distribution. Also explain the low-probability region near $x_1 = x_2$.

(ii) Compute

$$\langle |x_1 - x_2| \rangle \tag{11}$$

for at least $\lambda = 0.5, 1.0, 1.5, 2.0$, and plot it as a function of λ. Include the noninteracting reference point

$$\langle |x_1 - x_2| \rangle_{\lambda=0} = \frac{2L}{3} = \frac{2}{3}. \tag{12}$$

On the same graph, include (or otherwise directly compare with) the values you previously obtained using rejection sampling. The two methods should agree within statistical uncertainty. Briefly discuss the trend with increasing λ and any differences you observe in sampling efficiency or statistical noise.